

Acquired Cystic Kidney Disease and Malignant Neoplasms

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DEFINITION

Acquired cystic kidney disease (ACKD) was first recognized in 1847 by John Simon in patients with chronic Bright disease. He described the development of cystic kidney changes with cysts ranging from “mustard seed to as large as cocoa nuts” and also noted that they “run a slow and insidious progress during life, and often leave in the dead body no such obvious traces as would strike the superficial observer.” ACKD was “rediscovered” by Dunnill and colleagues¹ in 1977 in kidneys from dialysis patients.

ACKD is a specific entity of chronic kidney disease (CKD) of any etiology and has to be differentiated from other types of cystic kidney disease (see [Chapters 46 and 47](#)). It is usually defined as more than three macroscopic cysts in each kidney of a patient who does not have a hereditary cause of cystic disease. Some researchers consider ACKD to be preneoplastic.^{2,3}

PATHOGENESIS

The exact mechanisms of tubule transformation into cysts in ACKD are unknown. Responses to acute or chronic kidney injury with parenchymal (nephron) loss are believed to be involved. Progressive nephron loss results in initial tubular hypertrophy followed by hyperplasia and is likely aggravated by tubular distortion and distal tubular outflow obstruction, such as by advanced tubular atrophy, interstitial fibrosis, or calcium oxalate crystals, which are all common in ACKD ([Fig. 92.1](#)).⁴ ACKD can appear before dialysis is started and does not correlate with dialysis modality or underlying kidney disease. The cysts seem to develop from various tubular segments. Most cysts are lined not only by a single layer of flat to cuboidal epithelial cells but also by cuboidal cells with eosinophilic, foamy, and occasionally also clear cytoplasm, some of which are considered precursor lesions of renal cell carcinomas (RCC).^{2,3,5}

With the continuing presence of mitogenic stimuli, the intracystic epithelium becomes multilayered or forms micropapillary proliferates. Further accumulation of mutations, activated protooncogenes, or chromosomal abnormalities, in conjunction with additional factors such as genetic background, environmental chemicals, and sex hormones, most likely accounts for the transition to RCC (see [Fig. 92.1](#)).⁶ Next-generation sequencing revealed a gain of chromosome 16 and recurrent mutations in the *KMT2c* and *TSC2* genes, which might act as oncogenic drivers in the specific subtype of RCC found in ACKD (i.e., ACKD-associated RCC). So far, these mutations have no implications for therapy, but testing for these mutations in kidney tumors or cysts might in the future be developed into a tool for planning surveillance.⁷ ACKD-associated RCC also show a high expression of proximal tubular cell marker genes indicating this segment of the nephron as a potential origin.⁸

EPIDEMIOLOGY

Among adult and pediatric patients starting maintenance dialysis treatment, the prevalence of ACKD ranges from 5% to 20%. In both chronic hemodialysis (HD) and peritoneal dialysis (PD) patients, prevalence then increases at a similar rate and reaches 80% to 100% after 10 years of treatment ([Fig. 92.2](#)).^{9–11} The rate of progression appears to slow after 10 to 15 years of dialysis.

The frequency of ACKD and kidney tumors in dialysis patients may be underestimated based on imaging alone. Kidney cysts are detectable by ultrasound, with a minimum size of 0.5 cm. Data obtained in native nephrectomy specimens at the time of transplantation identified ACKD in 33%, kidney adenomas in 14%, and RCCs in 4% of the cases.¹² In unselected series of chronic dialysis or transplant patients, the cumulative incidence of RCC complicating ACKD as demonstrated by imaging is less than 1%, although rates up to 7% have been reported in some small studies. These data indicate an up to 40- to 100-fold increased risk for RCC in ACKD patients compared with RCC in the general population. Risk factors include male sex (male-to-female ratio, 7:1), Black ethnicity, long duration of dialysis, and severe ACKD with marked organ enlargement.

CLINICAL MANIFESTATIONS

ACKD can manifest as unilateral or bilateral cysts, which are mostly cortical and variable in size and number. Rarely, severe ACKD can become macroscopically indistinguishable from adult polycystic kidney disease (PKD). In contrast to hereditary cystic diseases, the cysts of ACKD are strictly confined to the kidneys. The disease is usually asymptomatic and discovered accidentally during abdominal imaging procedures. Alternatively, it may be manifested by the following potential complications or consequences of ACKD¹³:

- Cystic hemorrhage with or without hematuria; bleeding may occur with cyst rupture with subsequent perinephric hemorrhage or retroperitoneal hemorrhage, which can rarely lead to hypovolemic shock.
- Calcifications in or around cysts and in rare cases stone formation (calcium-containing stones or β_2 -microglobulin stones).
- Cyst infection, abscess formation, or sepsis.
- Erythrocytosis in advanced cases, similar to the erythrocytosis observed in PKD.
- Malignant transformation.

Acquired Cystic Kidney Disease: Associated Renal Cell Carcinomas

About 85% of RCCs in ACKD are asymptomatic at diagnosis. The remainder mostly manifest with bleeding, usually gross hematuria. In cases in which nephrectomy was done in dialysis patients for intractable

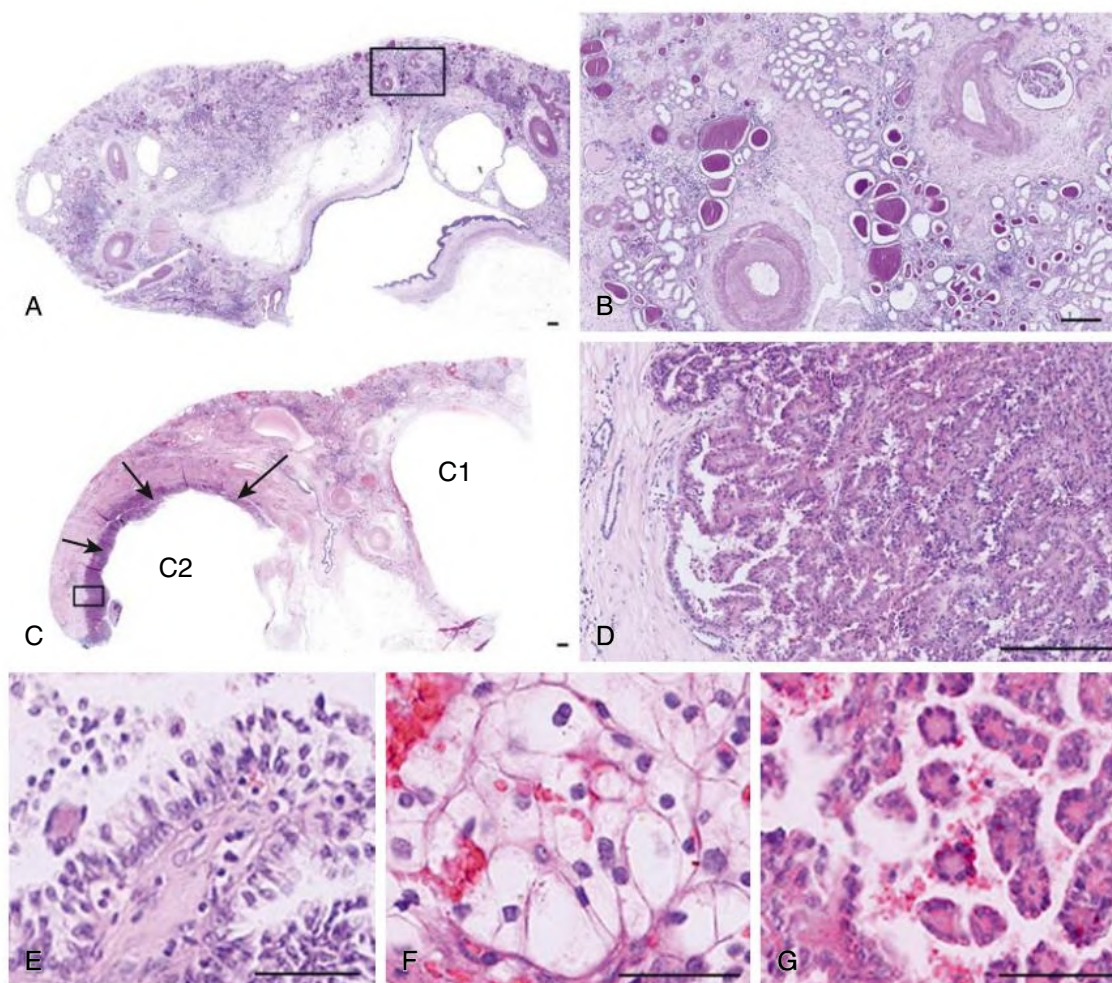


Fig. 92.1 Typical Histomorphologic Appearance of Acquired Cystic Kidney Disease (ACKD) and Commonly Found Renal Cell Carcinoma (RCC) in ACKD. (A) The kidney shows numerous cysts of different sizes and prominent atrophy of kidney parenchyma. (B) At higher magnification global glomerulosclerosis, tubular atrophy with typical focal areas of thyroidization (i.e., intratubular periodic acid–Schiff [PAS]-positive casts), interstitial fibrosis, and arteriosclerosis (PAS-stained section) are noted. (C) Another area of the same kidney specimen as shown in part A, with two cysts. The right cyst is lined by inconspicuous and at this magnification invisible single-layered epithelium (C1). The left cyst (C2) is lined by a thick cell-dense layer (arrows), which at higher magnification (D) is characterized by tubular and papillary proliferation of atypical epithelial cells, characteristic for intracystic ACKD-associated RCC (H&E stained section). (E) Another tumor often found in ACKD is composed of well-differentiated clear cells in a papillary arrangement, characteristic for clear cell papillary RCC. (F) A clear cell RCC, a tumor commonly found in ACKD and the most common kidney tumor in the general population. It is composed of nests and sheets of atypical cells with clear cytoplasm. (G) Papillary RCC, another tumor commonly found in ACKD and the second most common RCC in the general population. It is composed of eosinophilic epithelial cells in a papillary arrangement. In this case the tumor size was 3 cm; therefore the diagnosis of papillary RCC (type 1) was made; if the size were less than 1.5 cm, the same histologic tumor pattern would be classified as papillary adenoma. In the presented cases and typical for ACKD, most clear cell and papillary RCCs in ACKD are well-differentiated, that is, the nuclei show little or no nucleoli and little nuclear pleomorphism. The scale bars in A–D represent 250 μ m and in E–G represent 50 μ m. The insets in A and C represent areas shown in higher magnification in B and D, respectively.

hematuria, previously undetected RCCs were diagnosed in about one-third of the patients. Compared with sporadic RCCs, ACKD-associated RCCs are characterized by younger age of the patient, male predominance, more frequent multicentric and bilateral manifestation, and less frequent metastases.²

PATHOLOGY

Cystic changes in ACKD are typically bilateral but may vary between kidneys. Most ACKD kidneys are smaller than normal, and all show

advanced parenchymal atrophy and chronic injury. An increase in the size of the ACKD kidneys above normal may result from cyst hemorrhage or malignant transformation. The cysts are usually restricted to the kidney cortex. The size of the cysts ranges from microscopic to about 2 cm; about 60% of the cysts are smaller than 0.2 cm.² Several types of the epithelial lining of cysts can be observed, some of which are likely the earliest precursor lesions of RCCs in ACKD.¹⁴

Up to 25% of kidneys with ACKD harbor tumors when examined histologically. About one-quarter to one-third of these are carcinomas. Adenomas are only defined for well-differentiated papillary tumors and

Prevalence of Acquired Cystic Kidney Disease in Hemodialysis Patients

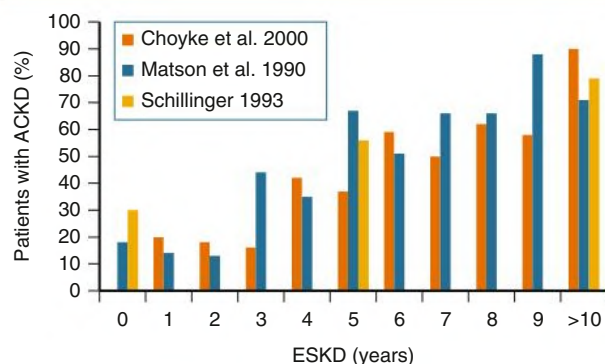


Fig. 92.2 Prevalence of Acquired Cystic Kidney Disease (ACKD) in Hemodialysis Patients. Summary of reported ACKD prevalence in chronic hemodialysis (HD) patients in relation to the length of HD treatment. ESKD, End-stage kidney disease. (Modified from Choyke PL. Acquired cystic kidney disease. *Eur Radiol.* 2000;10:1716–1721; Matson MA, Cohen EP. Acquired cystic kidney disease: occurrence, prevalence, and renal cancers. *Medicine (Baltimore).* 1990;69:217–226; and Schillinger F. Acquired cystic kidney disease in renal insufficiency: a multicentre study. Group of Nephrologists of the East of France. *Eur J Med.* 1993;2:457–460.)

are arbitrarily considered adenomas when less than 1.5 cm, whereas those greater than 1.5 cm are considered carcinomas, given the higher metastatic probability of larger tumors. All other tumors showing typical histology of RCCs regardless of size (e.g., invasive groups of clear cells) are diagnosed as RCCs.

The most common type of RCC that is specific for ACKD is ACKD-associated RCC, which accounts for about 36% of RCCs found in ACKD. The histologic appearance of ACKD-associated RCCs varies from tubular, papillary, acinar, microcystic, to solid. They are mostly composed of eosinophilic tumor cells with prominent nucleoli and frequent intratumoral calcium oxalate crystals (see Fig. 92.1).⁵ The second most common RCC (24%) that is relatively specific for ACKD (but can be also observed in non-ACKD kidneys) is clear cell papillary RCC, which shows papillary, tubular, or cystic arrangement of relatively well-differentiated clear cells.

Other commonly observed RCCs in ACKD are the same as those in the general population (i.e., papillary [18%], clear cell [10%], and chromophobe RCCs [6%]), with some studies showing that the majority of RCCs in ACKD are clear cell (45%), ACKD-associated (32%), and papillary (13%).¹⁵ RCCs arising from ACKD are multicentric in about 50% of cases and bilateral in about 10%. Multiple histologic subtypes may occur in the same kidney. Clear cell RCCs contain cells in solid, tubulocystic, or alveolar arrangements, whereas papillary RCCs show eosinophilic, papillary arranged tumor cells (see Fig. 92.1). In difficult cases with overlapping histologic appearance, immunohistochemistry helps differentiate between these entities. The precise diagnosis is important because ACKD-associated RCCs and clear cell papillary RCCs seem to be more indolent than clear cell RCCs but can also metastasize. All RCCs in ACKD tend to have a better clinical course most likely as a result of early detection given close follow-up of these patients. Compared with other RCCs, ACKD-associated RCCs less frequently showed unfavorable pathologic findings such as sarcomatoid differentiation, lymphovascular invasion, or necrosis, which might be other reasons for a more favorable prognosis.¹⁵

RCCs in native kidneys of kidney transplant patients tend to be smaller and exhibit a lower T stage and a lower grade at diagnosis

compared with RCCs in patients with end-stage kidney disease (ESKD). Tumors are more often multifocal, bilateral, and papillary.^{16,17} Likely because of the earlier stage at detection, the prognosis of RCCs after kidney transplantation is generally more favorable.

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

The diagnostic approach to ACKD usually involves ultrasound (Fig. 92.3).^{4,9} However, the differentiation between simple cysts and RCCs can be difficult given the echogenicity of end-stage kidney parenchyma and the complexity of cysts in ACKD. Computed tomography (CT) scanning, in particular with early contrast enhancement, is superior to ultrasound in detecting small malignant lesions (see Fig. 92.3).^{4,9,18} A classification of kidney cysts (Fig. 92.4) based on their appearance in CT scans, introduced by Bosniak,^{19,20} is now widely accepted and is also applied to ultrasound and magnetic resonance imaging (MRI; see Table 63.6). Criteria that favor the diagnosis of RCC as opposed to a simple cyst include thickened and irregular walls, the presence of septa or kidney tissue within the lesions, contrast enhancement, multilocularity, and large size (≥ 4 cm). MRI scan can be an alternative to contrast-based CT scans. Although the risk for nephrogenic fibrosis after exposure to gadolinium-containing contrast agents is not a concern anymore in early stages of CKD, caution is still recommended in dialysis patients (see Chapter 6). For these cases, a MRI protocol without contrast enhancement has been developed and has been shown in a pilot study in kidney transplant recipients to be superior to kidney ultrasound alone.²¹ Another small study found that ¹¹C-choline-positron emission tomography (PET)/CT was more sensitive to detect RCC in ACKD compared with fluorodeoxyglucose (FDG)-PET/CT and contrast-enhanced CT, but this method is not broadly available.²²

Because of the risk for malignant transformation, testing for ACKD and follow-up imaging when ACKD is detected have both been advocated (Fig. 92.5). However, the cost of testing has to be weighed against the risk-to-benefit ratio of nephrectomy in dialysis patients. A decision analysis²³ concluded that testing for ACKD (by either ultrasound or CT scanning) in young patients with a life expectancy of 25 years offers as much as a 1.6-year gain in life expectancy. In contrast, in ACKD patients older than 60 years, no significant gain in life expectancy is achieved by regular testing.²⁴ In a different analysis of 797 dialysis patients who had developed RCCs (90% identified by regular testing), testing provided a mean survival benefit of 3.3 years after adjustment for age and dialysis vintage.²⁴ Testing during transplant evaluation by ultrasound followed by CT in the case of suspicious lesions is recommended based on recent data showing a prevalence of kidney cancer in up to 4% of the patients and concerns about the role of immunosuppression in accelerating tumor growth.²⁵

NATURAL HISTORY

Cystic dilations of kidney tubules may develop at a glomerular filtration rate (GFR) lower than 70 mL/min.²⁶ ACKD thereafter progresses and reaches a prevalence of nearly 100% after more than 10 years of dialysis (see Fig. 92.2). The constant increase in kidney volume seems to reach a plateau after about 20 years of HD, and at least partial regression may occur after very-long-term HD.²⁷ In malignant transformation, tumor growth rates are highly variable. Death is usually associated with widespread metastases and accounts for about 2% of the deaths in kidney transplant patients.

It is not established whether kidney transplantation affects the natural history of RCC complicating ACKD, although immunosuppression has been suggested as a risk factor for RCC in transplant patients

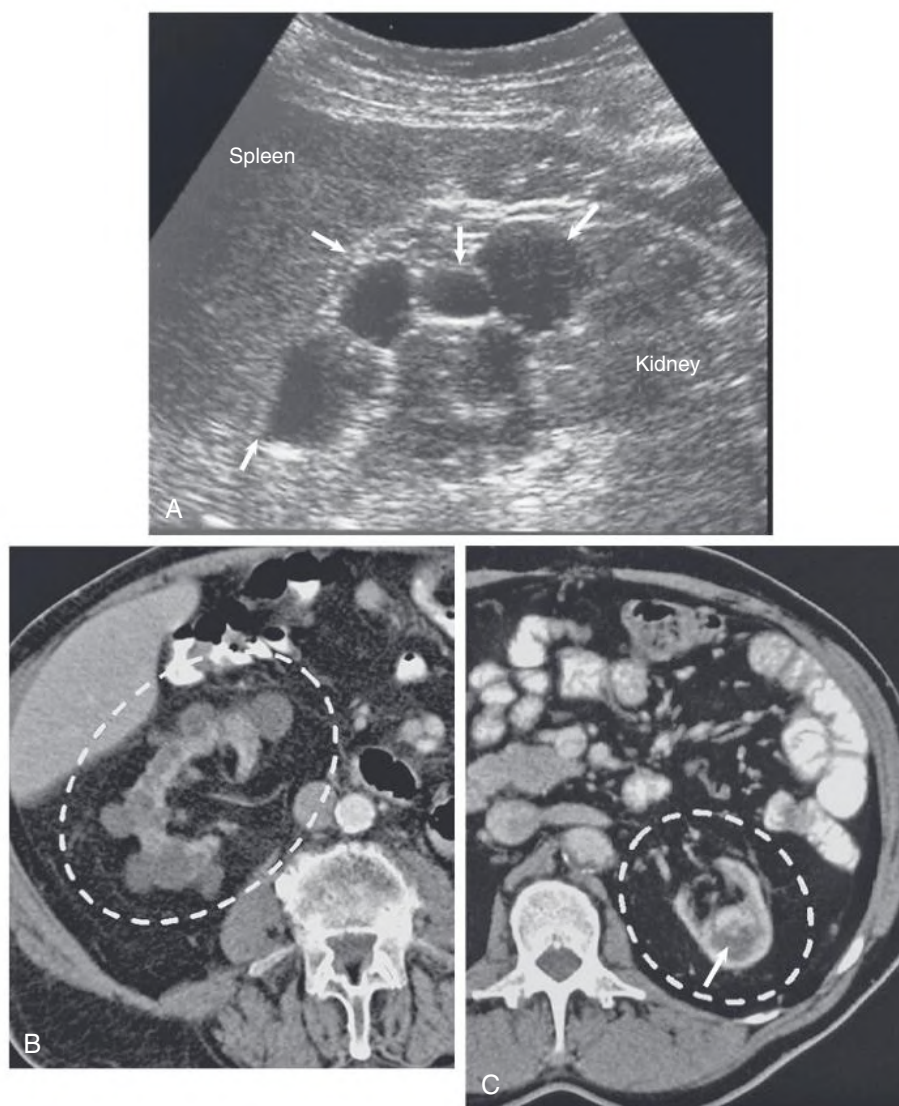


Fig. 92.3 Imaging Studies in Acquired Cystic Kidney Disease (ACKD). (A) Ultrasound image of the left native kidney of a patient after 16 years of chronic hemodialysis (HD). Multiple cysts are present in the kidney cortex (*arrows*). (B) Computed tomography (CT) image of a patient after 5 years of chronic HD demonstrating multiple cysts within the right kidney (*dashed circle*). (C) Contrast-enhanced CT image of a kidney transplant patient who developed a renal cell carcinoma (*arrow*) originating from the left native kidney with ACKD (*dashed circle*).

with ACKD.¹³ A prospective, single-center study in which 561 kidney transplant recipients were imaged with ultrasound identified ACKD in 23%.²⁸ The mean duration of dialysis was 4 to 5 years, and the mean time after transplantation was 9 years. In this cohort, ACKD was slightly less frequent than had been reported in dialysis patients, possibly indicating that kidney transplantation might inhibit the development of ACKD.²⁸ The prevalence of RCCs among all 561 patients was 4.8%. However, among patients with ACKD, RCCs were detected in almost 20%, but in only 0.5% of patients without ACKD.²⁸

TREATMENT

Treatment of ACKD is warranted only when complications such as hemorrhage, cyst infection, or malignant transformation develop. Although the first two complications may be handled conservatively and only rarely require surgery, malignant transformation should raise the question of nephrectomy. Given the perioperative morbidity and

mortality of nephrectomy, in particular in multimorbid dialysis or transplant patients, it is not surprising that the threshold for surgical intervention in cases of RCC is still controversial.

Most agree that tumors larger than 3 cm in diameter justify biopsy or surgical treatment (either tumor enucleation or nephrectomy) because, above this size, RCCs in the general population frequently metastasize (for details on urologic evaluation and treatment for kidney masses, refer to [Chapter 63](#)).¹¹ However, this strategy is based on an extrapolation from otherwise healthy persons, and a more aggressive approach may be required under certain circumstances. A Japanese study found that surgically treated RCCs in patients with ESKD seem to have a comparable stage-specific recurrence-free survival, but poorer cancer-specific and overall survival compared with non-ESKD RCC.²⁹

For tumors of less than 3 cm in diameter with no complications, the slow tumor growth may justify observation with repeated imaging studies. Patients with high life expectancy or listed for transplantation should

Testing for Renal Cell Carcinoma in Patients With Acquired Cystic Kidney Disease

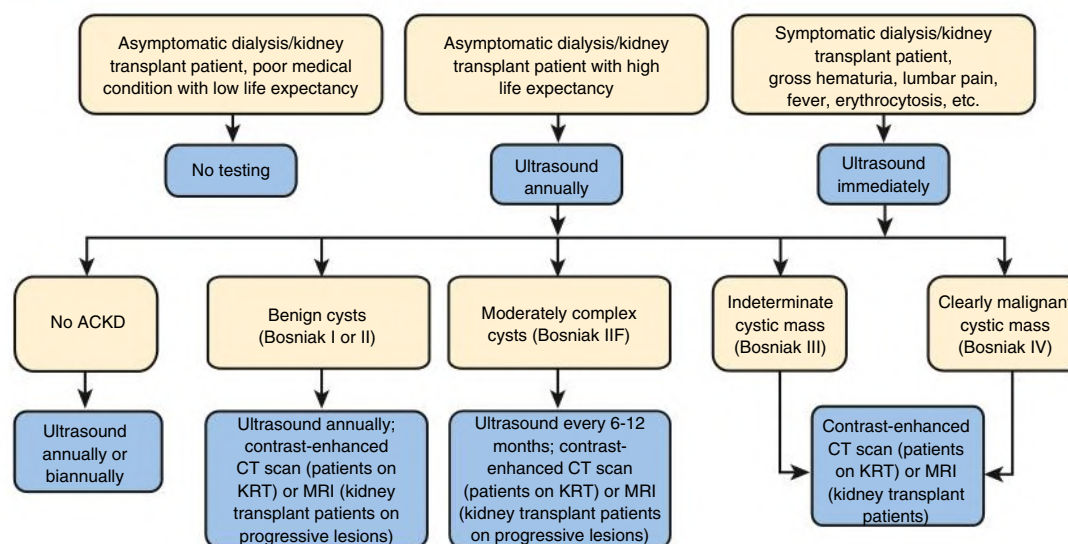


Fig. 92.4 Proposed Approach to Testing for Renal Cell Carcinoma in Patients with Acquired Cystic Kidney Disease (ACKD). *CT*, Computed tomography; *MRI*, magnetic resonance imaging; *KRT*, kidney replacement therapy. (Modified from Truong LD, Krishnan B, Cao JT, et al. Renal neoplasm in acquired cystic kidney disease. *Am J Kidney Dis.* 1995;26:1–12; Sarasin FP, Wong JB, Levey AS, Meyer KB. Screening for acquired cystic kidney disease: a decision analytic perspective. *Kidney Int.* 1995;48:207–219; and Schwarz A, Vatan-daslar S, Merkel S, Haller H. Renal cell carcinoma in transplant recipients with acquired cystic kidney disease. *Clin J Am Soc Nephrol.* 2007;2:750–756.)

Cancer Risk in Dialysis Patients

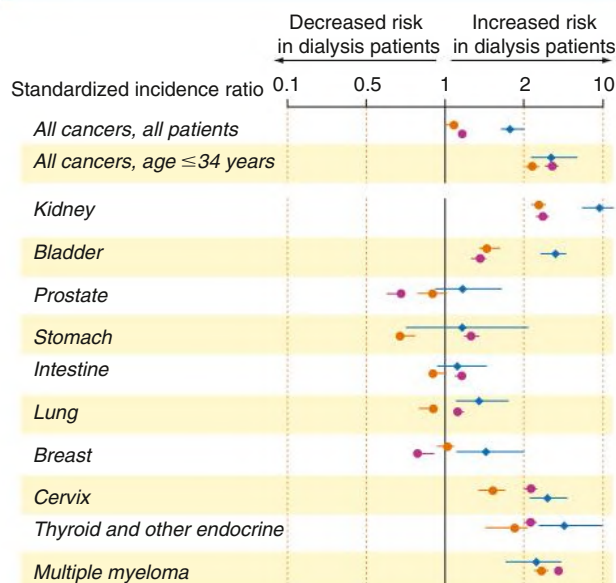


Fig. 92.5 Cancer Risk in Dialysis Patients. Relative risk for cancer (plus 95% confidence interval) compared with the general populations in 831,804 dialysis patients from Australia/New Zealand (blue diamonds), Europe (orange circles), and the United States (purple circles). (Modified from Maisonneuve P, Agodoa L, Gellert R, et al. Cancer in patients on dialysis for end-stage renal disease: an international collaborative study. *Lancet.* 1999;354:93–99.)

be considered for nephrectomy also in case of tumors with diameters less than 3 cm. In general, tumor enlargement should be used as an indication for nephrectomy if it is permitted by the patient's status.

In patients with unilateral tumors, prophylactic contralateral nephrectomy is not recommended because of the morbidity associated with the procedure, the worsening of anemia, and the loss of residual kidney function in those who are not considered transplant candidates. A delay of transplantation is not typically required in patients who have had an asymptomatic RCC associated with ACKD in their nephrectomy.

We recommend imaging transplant recipients yearly using ultrasound. In suspicious cases, contrast-enhanced CT scanning or MRI might help differentiate blood-filled cysts from solid tumors. In the case of a suspicious mass or RCC in the native kidneys, a nephrectomy should be performed. However, although immunosuppression is possibly associated with the development of RCC, it does not seem to diminish the likelihood of cure, especially because RCCs in transplant patients tend to be detected at an earlier stage, possibly due to frequent imaging in this population.³⁰

CANCER IN DIALYSIS PATIENTS

Dialysis patients have a slightly higher cancer risk compared with the general population. Analysis of more than 800,000 dialysis patients in three registries from the United States, Europe, and Australia/New Zealand revealed that most of the increased risk was because of cancers of the kidney, bladder, and endocrine organs (see Fig. 92.5).³¹ Besides the specific risk associated with malignant transformation of ACKD, some of the increased risk is directly related to the

underlying kidney disease or to the immunosuppression that may have been administered to patients with immune-mediated kidney disease. For example, cyclophosphamide may predispose to bladder and ureteral cancer that manifests after patients have been initiated on dialysis. Kidney disease or immunosuppressive therapy may underlie the apparent risk in dialysis patients for the development of multiple myeloma (see Fig. 92.5). In addition, patients with analgesic

nephropathy or aristolochic acid nephropathy are at high risk for the development of urothelial carcinoma of the upper urinary tract.³² Other malignancies that are more common in dialysis patients include carcinoma of the cervix, thyroid, and other endocrine neoplasias (see Fig. 92.5), and, at least in the U.S. Renal Data System database, a 1.5- to two fold increase in the risk for non-Hodgkin lymphoma, Hodgkin disease, and leukemias.³¹

SELF-ASSESSMENT QUESTIONS

1. A 52-year-old White man comes to your clinic with new-onset hematuria. He is receiving chronic hemodialysis (HD) treatment for ESKD because of diabetic nephropathy. Ultrasound of his kidneys reveals four cysts on the left side and five cysts on the right. On the right side, two of the cysts are not echo free and have a thickened wall with multiple calcifications. Which of the following statements is *correct*?
 - A. His previous diagnosis of diabetic nephropathy causing ESKD is false. Kidney ultrasound clearly depicts the typical picture of autosomal dominant polycystic kidney disease.
 - B. ACKD is a common finding in patients on maintenance HD treatment and is a possible cause of hematuria.
 - C. This patient needs to be carefully evaluated for the presence of RCC because kidney cysts in patients on RRT almost always undergo malignant transformation.
 - D. Most patients with ACKD have cystic disease outside of the kidneys (liver, pancreas) and need to be screened for cerebral aneurysms.
 - E. ACKD is associated with exposure to toxic compounds such as benzidine and 2-naphthylamine.
2. A 45-year-old Black woman on maintenance HD for lupus nephritis is asking for your advice. She has heard about an increased risk for malignant tumors in HD patients. Which of the following statements is *false*?
 - A. The increase in risk for tumor development in patients on RRT is, in part, related to the underlying kidney disease or to the immunosuppression that may have been administered to patients with immune-mediated kidney disease.
 - B. Most of the increased risk is because of cancers of the kidney, bladder, and endocrine organs.
 - C. Cyclophosphamide therapy may predispose to cancer of the bladder.
 - D. Patients with analgesic nephropathy or Chinese herbs/aristolochic acid nephropathy are at increased risk for development of transitional cell carcinoma (TCC) of the upper urinary tract.
 - E. Data from large registries reveal a nearly 10-fold increase in the risk for colorectal cancer in patients on HD treatment.
3. A kidney transplant recipient with ESKD secondary to focal segmental glomerulosclerosis presents for a regular checkup. He reports that his primary care physician had ordered an abdominal ultrasound because of recurrent right upper quadrant pain. As an incidental finding, multiple cysts in his native kidneys were noted. Which of the following statements about his condition is *false*?
 - A. ACKD is a common finding in patients with ESKD.
 - B. Up to 25% of kidneys with ACKD harbor tumors, about one-third of which are carcinomas.
 - C. ACKD is mostly asymptomatic; however, clinical manifestations include cyst hemorrhage, cyst infection, and malignant transformation.
 - D. Bilateral nephrectomy is the treatment of choice for asymptomatic ACKD in kidney transplant recipients because malignant transformation occurs in almost 90% of cases.
 - E. Contrast-enhanced computed tomography (CT) and magnetic resonance imaging (MRI) are the methods of choice to detect small malignant lesions in patients on RRT.

REFERENCES

- Dunnill MS, Millard PR, Oliver D. Acquired cystic disease of the kidneys: a hazard of long-term intermittent maintenance haemodialysis. *J Clin Pathol.* 1977;30:868–877.
- Truong LD, Choi YJ, Shen SS, et al. Renal cystic neoplasms and renal neoplasms associated with cystic renal diseases: pathogenetic and molecular links. *Adv Anat Pathol.* 2003;10:135–159.
- Hosseini M, Antic T, Paner GP, et al. Pathologic spectrum of cysts in end-stage kidneys: possible precursors to renal neoplasia. *Hum Pathol.* 2014;45:1406–1413.
- Nadasdy T, Laszik Z, Lajoie G, et al. Proliferative activity of cyst epithelium in human renal cystic diseases. *J Am Soc Nephrol.* 1995;5:1462–1468.
- Tickoo S, de Peralta-Venturina M, Harik L, et al. Spectrum of epithelial neoplasms in end-stage renal disease: an experience from 66 tumor-bearing kidneys with emphasis on histologic patterns distinct from those in sporadic adult renal neoplasia. *Am J Surg Pathol.* 2006;30:141–153.
- Grantham JJ. Acquired cystic kidney disease. *Kidney Int.* 1991;40:143–152.
- Shah A, Lal P, Toorens E, Palmer MB, et al. Acquired cystic kidney disease-associated renal cell carcinoma (ACKD-RCC) harbor recurrent mutations in KMT2C and TSC2 genes. *Am J Surg Pathol.* 2020;44(11):1479–1486.
- Ishihara H, Yamashita S, Liu YY, et al. Genetic and epigenetic profiling indicates the proximal tubule origin of renal cancers in end-stage renal disease. *Cancer Sci.* 2020;111(11):4276–4287.
- Choyke PL. Acquired cystic kidney disease. *Eur Radiol.* 2000;10:1716–1721.
- Matson MA, Cohen EP. Acquired cystic kidney disease: occurrence, prevalence, and renal cancers. *Medicine (Baltim).* 1990;69:217–226.
- Schillinger F. Acquired cystic kidney disease in renal insufficiency: a multicentre study. Group of Nephrologists of the East of France. *Eur J Med.* 1993;2:457–460.
- Denton MD, Magee CC, Ovuworie C, et al. Prevalence of renal cell carcinoma in patients with ESRD pre-transplantation: a pathologic analysis. *Kidney Int.* 2002;61:2201–2209.
- Truong LD, Krishnan B, Cao JT, et al. Renal neoplasm in acquired cystic kidney disease. *Am J Kidney Dis.* 1995;26:1–12.
- Sun Y, Argani P, Tickoo SK, et al. Renal cell carcinoma (ACKD-RCC)-like cysts. *Am J Surg Pathol.* 2018;42(10):1396–1401.
- Kondo T, Sasa N, Yamada H, et al. Acquired cystic disease-associated renal cell carcinoma is the most common subtype in long-term dialyzed patients: central pathology results according to the 2016 WHO classification in a multi-institutional study. *Pathol Int.* 2018;68(10):543–549.
- Klatte T, Seitz C, Waldert M, et al. Features and outcomes of renal cell carcinoma of native kidneys in renal transplant recipients. *BJU Int.* 2009;105:1260–1265.
- Gigante M, Neuzillet Y, Patard J, et al. Renal cell carcinoma (RCC) arising in native kidneys of dialyzed and transplant patients: are they different entities?. *BJU Int.* 2012 (epub ahead of print).
- Takebayashi S, Hidai H, Chiba T, et al. Using helical CT to evaluate renal cell carcinoma in patients undergoing hemodialysis: value of early enhanced images. *AJR Am J Roentgenol.* 1999;172:429–433.
- Israel GM, Bosniak MA. An update of the Bosniak renal cyst classification system. *Urology.* 2005;66:484–488; and modified according to Eknayan G. A clinical view of simple and complex renal cysts. *J Am Soc Nephrol.* 2009;20:1874–1876.
- Israel GM, Bosniak MA. An update of the Bosniak renal cyst classification system. *Urology.* 2005;66:484–488.
- Mühlfeld AS, Lange C, Kroll G, et al. Pilot study of non-contrast-enhanced MRI vs. ultrasound in renal transplant recipients with acquired cystic kidney disease: a prospective intra-individual comparison. *Clin Transplant.* 2013;27(6):E694–E701.
- Kitajima K, Yamamoto S, Kawanaka Y, et al. Imaging of renal cell carcinoma in patients with acquired cystic disease of the kidney: comparison 11 C-choline and FDG PET/CT with dynamic contrast-enhanced CT. *Jpn J Radiol.* 2019;37(2):165–177.
- Sarasin FP, Wong JB, Levey AS, Meyer KB. Screening for acquired cystic kidney disease: a decision analytic perspective. *Kidney Int.* 1995;48:207–219.
- Ishikawa I, Honda R, Yamada Y, Kakuma T. Renal cell carcinoma detected by screening shows better patient survival than that detected following symptoms in dialysis patients. *Ther Apher Dial.* 2004;8:468–473.
- Gulanikar AC, Daily PP, Kilambi NK, et al. Prospective pretransplant ultrasound screening in 206 patients for acquired renal cysts and renal cell carcinoma. *Transplantation.* 1998;66:1669–1672.
- Liu JS, Ishikawa I, Horiguchi T. Incidence of acquired renal cysts in biopsy specimens. *Nephron.* 2000;84:142–147.
- Ishikawa I, Hayama S, Morita K, et al. Long-term natural history of acquired cystic disease of the kidney. *Ther Apher Dial.* 2010;14:409–416.
- Schwarz A, Vatasdaslar S, Merkel S, Haller H. Renal cell carcinoma in transplant recipients with acquired cystic kidney disease. *Clin J Am Soc Nephrol.* 2007;2:750–756.
- Song C, Hong SH, Chung JS, et al. Renal cell carcinoma in end-stage renal disease: multi-institutional comparative analysis of survival. *Int J Urol.* 2016;23(6):465–471.
- Tollefson M, Krambeck A, Leibovich B, et al. Surgical treatment of renal cell carcinoma in the immunocompromised transplant patient. *Urology.* 2010;75:1373–1377.
- Maisonneuve P, Agodoa L, Gellert R, et al. Cancer in patients on dialysis for end-stage renal disease: an international collaborative study. *Lancet.* 1999;354:93–99.
- Stewart JH, Buccianti G, Agodoa L, et al. Cancers of the kidney and urinary tract in patients on dialysis for end-stage renal disease: analysis of data from the United States, Europe, and Australia and New Zealand. *J Am Soc Nephrol.* 2003;14:197–207.